LAPORAN UTS PPLOS A
SISTEM RESERVASI KLINIK / MANAJEMEN PASIEN

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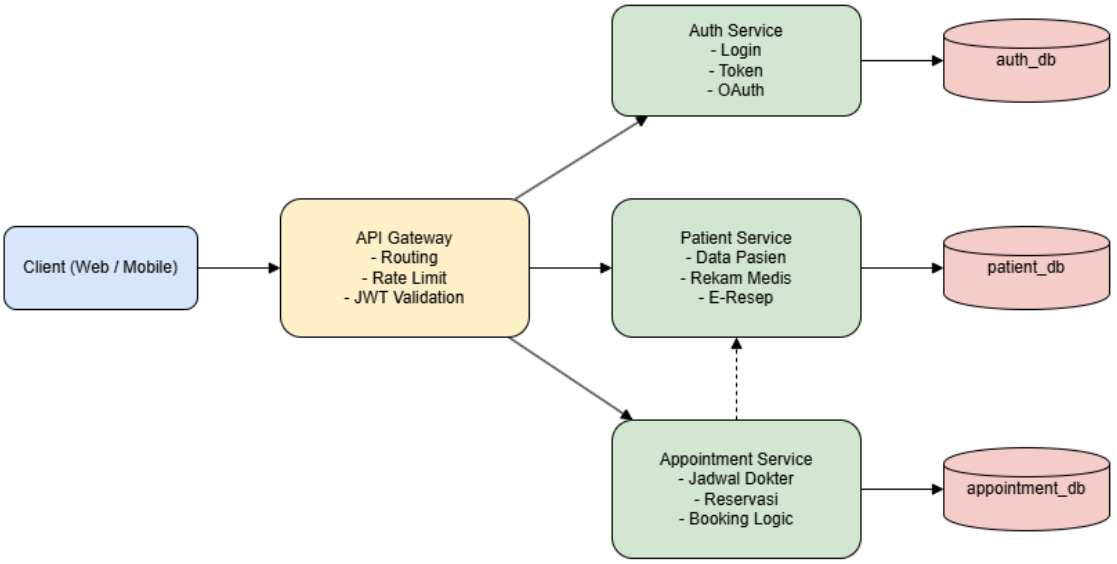
Studi Kasus : Sistem Reservasi Klinik / Manajemen Pasien
OAuth Wajib : GitHub OAuth 2.0 Authorization Code Flow

A. Arsitektur (Microservice System)



Sistem terdiri dari 4 komponen utama:

- API Gateway (Node.js + Express, Port 3000): Single entry point untuk routing dan validasi JWT.
- Auth Service (Node.js + Express, Port 3001): Mengelola autentikasi dan otorisasi user.
- Patient Service (PHP MVC dengan CodeIgniter 4, Port 8000): Mengelola data pasien, rekam medis, dan resep.
- Appointment Service (Node.js + Express, Port 3002): Mengelola jadwal dokter dan reservasi online.

Setiap service memiliki database terpisah:

- auth_db untuk Auth Service
- patient_db untuk Patient Service
- appointment_db untuk Appointment Service

Pendekatan monolitik akan membuat seluruh aplikasi dalam satu codebase, yang sulit di-maintain dan scale. Dengan microservices:

- Auth Service dipisah untuk fokus pada security dan authentication, memungkinkan reuse di service lain tanpa duplikasi kode.

- Patient Service (PHP) dipisah karena menggunakan stack berbeda (PHP MVC), dan dikonsumsi oleh Appointment Service untuk validasi pasien saat reservasi.
- Appointment Service dipisah untuk mengelola logika bisnis reservasi secara independen, menghindari coupling dengan data pasien.
- API Gateway sebagai entry point tunggal, memungkinkan load balancing, rate limiting, dan centralized authentication tanpa service lain perlu mengetahui detail internal.

Semua endpoint menggunakan HTTP methods secara semantik:

- GET untuk retrieve data
- POST untuk create
- PATCH untuk update partial
- DELETE untuk remove

Response codes yang digunakan: 200 OK, 201 Created, 204 No Content, 400 Bad Request, 401 Unauthorized, 403 Forbidden, 404 Not Found, 409 Conflict, 422 Unprocessable Entity, 500 Internal Server Error.

B. REST API + Database (Relational Database)

Patient Service menggunakan MySQL dengan 4 tabel ter-relasi:

- patients (id, user_id, nik, name, birth_date, gender, phone, address)
- medical_records (id, patient_id, doctor_name, diagnosis, treatment, visit_date)
- prescriptions (id, record_id, medication, dosage, instructions)
- Relasi: patients → medical_records → prescriptions (one-to-many)

Auth Service menggunakan MySQL dengan 4 tabel ter-relasi:

- users (id, name, email, password, avatar_url, oauth_provider, oauth_id, role_id)
- roles (id, role_name)
- refresh_tokens (id, user_id, token_hash, expires_at, revoked_at)
- token_blacklist (id, token_hash, expires_at)
- Relasi: users → roles (many-to-one), users → refresh_tokens (one-to-many), users → token_blacklist (one-to-many)

Validasi dilakukan di layer controller:

- Patient Service: Validasi required fields, format date (Y-m-d), email, dll.
- Auth Service: Validasi email format, password length.
- Appointment Service: Validasi patient_id, schedule_id, dll.

Endpoint GET /patients menerapkan paging dengan query params page dan per_page, serta filtering berdasarkan name dan nik.

C. REST API PHP MVC

Patient Service dibangun menggunakan CodeIgniter 4 dengan pola MVC jelas:

- Model: PatientModel.php, MedicalRecordModel.php, PrescriptionModel.php - mengelola database operations.
- Controller: Patient.php, MedicalRecord.php, Prescription.php - menangani request/response dan validasi.

- Routes: Routes.php - mendefinisikan endpoint mapping.

Patient Service dikonsumsi oleh Appointment Service saat POST /reservations untuk memverifikasi apakah patient_id valid dengan memanggil GET /patients/{id} ke Patient Service.

D. Otentikasi JWT

Implementasi JWT

- Login dengan email + password menghasilkan access token (JWT, expires 15 menit) dan refresh token (expires 7 hari).
- Access token disimpan di header Authorization: Bearer <token>.
- Endpoint POST /auth/refresh untuk renew token menggunakan refresh token.
- Endpoint POST /auth/logout untuk invalidate token dengan blacklist access token dan revoke refresh token.

Gateway menggunakan middleware authenticateGateway untuk verifikasi JWT pada semua endpoint protected sebelum proxy ke service belakang.

E. Otorisasi OAuth 2.0

Implementasi GitHub OAuth menggunakan Authorization Code Flow:

- GET /auth/github redirect ke GitHub authorize URL.
- GET /auth/github/callback menukar code ke access token GitHub, fetch profil, dan map ke user lokal.
- Jika user belum ada, dibuat otomatis dengan oauth_provider = 'github'.
- Menyimpan: nama, email, foto profil dari GitHub.

F. API Gateway

Gateway sebagai single entry point:

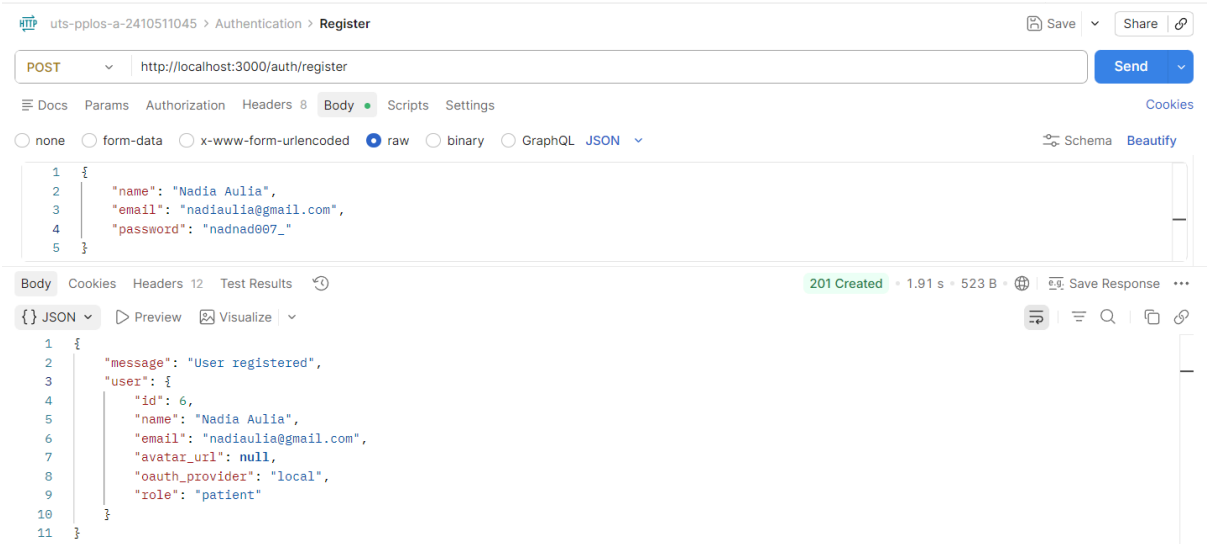
- Routing: Proxy request berdasarkan path ke service tujuan.
- JWT Validation: Middleware verifikasi token sebelum forward.
- Rate Limiting: 60 req/menit per IP menggunakan basic counter.

Peta Routing

Method	Endpoint Gateway	Service Tujuan	Proteksi
POST	/auth/register	Auth Service	Public
POST	/auth/login	Auth Service	Public
GET	/auth/github	Auth Service	Public
GET	/auth/github/callback	Auth Service	Public
POST	/auth/refresh	Auth Service	Public
POST	/auth/logout	Auth Service	JWT
POST	/auth/introspect	Auth Service	JWT/Internal

GET	/patients?page=1&per_page=10&name=&nik=	Patient Service	JWT
POST	/patients	Patient Service	JWT
GET	/records/:id	Patient Service	JWT
POST	/records	Patient Service	JWT
POST	/prescriptions	Patient Service	JWT
GET	/doctors/schedule	Appointment Service	JWT
POST	/reservations	Appointment Service	JWT
GET	/reservations/me	Appointment Service	JWT
PATCH	/reservations/:id/status	Appointment Service	JWT
DELETE	/reservations/:id	Appointment Service	JWT

Klien hanya mengakses melalui http://localhost:3000.



uts-pplos-a-2410511045 > Authentication > **Login**

Save | Share

POST http://localhost:3000/auth/login **Send**

Docs | Params | Authorization | Headers 8 | Body | Scripts | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

```
1 {
2     "email": "nadiaulia@gmail.com",
3     "password": "nadnad007_"
4 }
```

Body | Cookies | Headers 12 | Test Results | 200 OK | 1.43 s | 956 B | Save Response

{} JSON | Preview | Visualize


```
1 {
2     "token_type": "Bearer",
3     "expires_in": 900,
4     "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.
        eyJzdWIiOjYsImVtYWlsIjoibmFkaWF1bGlhQGdtYWlsLmNvbSIsIm5hbWUiOiJOYWRpYSBBdWxpYSIsInJvbGUiOiJwYXRpZW50IiwiaWF0IjoxNzc3NzQ2MDM2LCJleHAiOjE3Nz
        c3NDY5MzZ9.nDG2TwU9CkBWOR9xKQB54X9E4fq-JXxkfE_27xvls54",
5     "refresh_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOjYsInR5cGUiOiJyZWZyZXNoIiwiaWF0IjoxNzc3NzQ2MDM2LCJleHAiOjE3NzgzNTA4MzZ9.
        yBrLJun4f5btwwk5TYHkFuIGhRKCJ4SdaGpXN_k5pNw",
6     "user": {
7         "id": 6,
8         "name": "Nadia Aulia",
9         "email": "nadiaulia@gmail.com",
10        "avatar_url": null,
11        "oauth_provider": "local",
12        "role": "patient"
13    }
14 }
```


localhost:3000/auth/github/callback?code=f471ac1cb97c84b41066&state=oe3zvb80vta

Pretty-print


```
{"token_type":"Bearer","expires_in":900,"access_token":"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOjIsImVtYWlsIjoibmRhanNtbmVAZ21haWwuY29tIiwibmFtZSI6Ik5hZGlhIEphc21pbmUgQXVsaWEiLCJyb2xlIjoicGF0aWVudCIsImlhdCI6
MTc3Nzc0NTc4NSwiZXhwIjoxNzc3NzQ2Njg1fQ.akoJHJ51ZxJ-
9AZXctgNqg3iS_HUvnP9Mh7vLhjWQXk","refresh_token":"eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOjIsInR5cGUiOiJyZWZyZXNoIiwiaWF0IjoxNzc3NzQ1NzgiLCJleHAiOjE3NzgzNTA1ODV9.fLV2hrP0o18IhRG2jwiyWpWA32qkFdbCCaWIqMHPFZA",
"user":{"id":2,"name":"Nadia Jasmine Aulia","email":"ndajsmne@gmail.com","avatar_url":"https://avatars.githubusercontent.com/u/24314587?v=4","oauth_provider":"github","role":"patient"}}
```


uts-pplos-a-2410511045 > Authentication > **Refresh**

Save | Share

POST http://localhost:3000/auth/refresh **Send**

Docs | Params | Authorization | Headers 8 | Body | Scripts | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

```
1 {
2     "refresh_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOjYsInR5cGUiOiJyZWZyZXNoIiwiaWF0IjoxNzc3NzQ1NzQwLCJleHAiOjE3NzgzNTA1NDB9.
        DJpsENa_zM6PT6gr2Clx62Kk_UI0DhFB33QvwK5GFg4"
3 }
```

Body | Cookies | Headers 12 | Test Results | 200 OK | 4.89 s | 957 B | Save Response

{} JSON | Preview | Visualize


```
1 {
2     "token_type": "Bearer",
3     "expires_in": 900,
4     "access_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.
        eyJzdWIiOjYsImVtYWlsIjoibmFkaWF1bGlhQGdtYWlsLmNvbSIsIm5hbWUiOiJOYWRpYSBBdWxpYSIsInJvbGUiOiJwYXRpZW50IiwiaWF0IjoxNzc3NzQ1ODg1LCJleHAiOjE3Nz
        c3NDY3ODV9.bIbfeJBlmQM-XJ-d1p1fvXuUkxdB6DnqnKX_tUPcwoM",
5     "refresh_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOjYsInR5cGUiOiJyZWZyZXNoIiwiaWF0IjoxNzc3NzQ1ODg1LCJleHAiOjE3NzgzNTA2ODV9.
        oqYXjVha0nt2nuPFTqQut2ef6TWp94ju9Lny4Ir14uk",
6     "user": {
7         "id": 6,
8         "name": "Nadia Aulia",
9         "email": "nadiaulia@gmail.com",
10        "avatar_url": null,
11        "oauth_provider": "local",
12        "role": "patient"
13    }
14 }
```


uts-pplos-a-2410511045 > Authentication > **Logout**

Save | Share

POST http://localhost:3000/auth/logout **Send**

Docs | Params | Authorization | Headers 9 | Body | Scripts | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

```
1 {
2     "refresh_token": "eyJhbGciOiJIUzI1NiIsInR5cCI6IkpXVCJ9.eyJzdWIiOjYsInR5cGUiOiJyZWZyZXNoIiwiaWF0IjoxNzc3NzQ1ODg1LCJleHAiOjE3NzgzNTA2ODV9.
        oqYXjVha0nt2nuPFTqQut2ef6TWp94ju9Lny4Ir14uk"
3 }
```

Body | Cookies | Headers 9 | Test Results | 204 No Content | 999 ms | 260 B | Save Response

Raw | Preview | Visualize

```
1
```

uts-pplos-a-2410511045 > Authentication > **Introspect**

Save | Share

POST http://localhost:3000/auth/introspect Send

Docs | Params | Authorization | Headers 8 | Body | Scripts | Settings | Cookies

Auth Type: Bearer Token

Token: ʿJ-d1p1fvXuUkxdB6DnqnKX_tUPcwoM

The authorization header will be automatically generated when you send the request. Learn more about Bearer Token authorization.

Body | Cookies | Headers 12 | Test Results

401 Unauthorized · 688 ms · 413 B | Save Response

{} JSON | Preview | Pass the correct auth credentials


```
{
    "message": "Token is inactive or revoked"
}
```


uts-pplos-a-2410511045 > Appointment > **Lihat Jadwal Dokter**

Save | Share

GET http://localhost:3000/doctors/schedule Send

Docs | Params | Authorization | Headers 7 | Body | Scripts | Settings | Cookies

Body | Cookies | Headers 12 | Test Results

200 OK · 1.67 s · 783 B | Save Response

{} JSON | Preview | Visualize


```
{
    "data": [
        {
            "id": 1,
            "day_of_week": "Senin",
            "start_time": "08:00:00",
            "end_time": "12:00:00",
            "quota": 12,
            "doctor_id": 1,
            "doctor_name": "dr. Rania Putri",
            "specialization": "Dokter Umum",
            "clinic_id": 1,
            "clinic_name": "Poli Umum"
        },
        {
            "id": 2,
            "day_of_week": "Rabu",
            "start_time": "13:00:00",
            "end_time": "16:00:00",
            "quota": 8,
            "doctor_id": 2,
            "doctor_name": "drg. Arif Maulana",
            "specialization": "Dokter Gigi",
            "clinic_id": 2,
            "clinic_name": "Poli Gigi"
        }
    ]
}
```

uts-pplos-a-2410511045 > Appointment > **Buat Reservasi**

Save | Share

POST http://localhost:3000/reservations **Send**

Docs | Params | Authorization | Headers 9 | Body | Scripts | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

```
1  {
2    "patient_id": 1,
3    "schedule_id": 1,
4    "reservation_date": "2026-05-12",
5    "complaint": "Kontrol kesehatan umum"
6  }
```

Body | Cookies | Headers 12 | Test Results | 201 Created | 5.31 s | 768 B | Save Response

{} JSON | Preview | Visualize

```
1  {
2    "message": "Reservation created",
3    "data": {
4      "id": 2,
5      "user_id": 6,
6      "patient_id": 1,
7      "schedule_id": 1,
8      "reservation_date": "2026-05-11T17:00:00.000Z",
9      "complaint": "Kontrol kesehatan umum",
10     "status": "pending",
11     "created_at": "2026-05-02T11:22:01.000Z",
12     "updated_at": "2026-05-02T11:22:01.000Z",
13     "day_of_week": "Senin",
14     "start_time": "08:00:00",
15     "end_time": "12:00:00",
16     "doctor_name": "dr. Rania Putri",
17     "clinic_name": "Poli Umum"
18   }
19 }
```

uts-pplos-a-2410511045 > Appointment > **Lihat Detail Reservasi**

Save | Share

GET http://localhost:3000/reservations/me **Send**

Docs | Params | Authorization | Headers 7 | Body | Scripts | Settings | Cookies

Auth Type: Bearer Token

Token: VOR9xKQB54X9E4fq-JXxkfE_27xvls54

The authorization header will be automatically generated when you send the request. Learn more about Bearer Token authorization.

Body | Cookies | Headers 12 | Test Results | 200 OK | 1.02 s | 733 B | Save Response

{} JSON | Preview | Visualize

```
1  {
2    "data": [
3      {
4        "id": 2,
5        "user_id": 6,
6        "patient_id": 1,
7        "schedule_id": 1,
8        "reservation_date": "2026-05-11T17:00:00.000Z",
9        "complaint": "Kontrol kesehatan umum",
10       "status": "pending",
11       "created_at": "2026-05-02T11:22:01.000Z",
12       "updated_at": "2026-05-02T11:22:01.000Z",
13       "day_of_week": "Senin",
14       "start_time": "08:00:00",
15       "end_time": "12:00:00",
16       "doctor_name": "dr. Rania Putri",
17       "clinic_name": "Poli Umum"
18     }
19   ]
20 }
```

uts-pplos-a-2410511045 > Appointment > **Update Status Reservasi**

Save | Share

PATCH http://localhost:3000/reservations/2/status Send

Docs Params Authorization Headers 9 Body Scripts Settings Cookies

none form-data x-www-form-urlencoded raw binary GraphQL JSON Schema Beautify

```
{
    "status": "confirmed"
}
```

Body Cookies Headers 12 Test Results 200 OK 313 ms 771 B Save Response

{} JSON Preview Visualize


```
{
    "message": "Reservation status updated",
    "data": {
        "id": 2,
        "user_id": 6,
        "patient_id": 1,
        "schedule_id": 1,
        "reservation_date": "2026-05-11T17:00:00.000Z",
        "complaint": "Kontrol kesehatan umum",
        "status": "confirmed",
        "created_at": "2026-05-02T11:22:01.000Z",
        "updated_at": "2026-05-02T11:23:33.000Z",
        "day_of_week": "Senin",
        "start_time": "08:00:00",
        "end_time": "12:00:00",
        "doctor_name": "dr. Rania Putri",
        "clinic_name": "Poli Umum"
    }
}
```


uts-pplos-a-2410511045 > Appointment > **Hapus Reservasi**

Save | Share

DELETE http://localhost:3000/reservations/2 Send

Docs Params Authorization Headers 7 Body Scripts Settings Cookies

Auth Type

Bearer Token

Token VOR9xKQB54X9E4fq-JXxkfE_27xvls54

The authorization header will be automatically generated when you send the request. Learn more about Bearer Token authorization.

Body Cookies Headers 9 Test Results 204 No Content 1.66 s 260 B Save Response

Raw Preview Visualize

```
1
```

uts-pplos-a-2410511045 > Patient > **Lihat Pasien**

Save | Share

GET http://localhost:3000/patients?page=1&per_page=10&name=Angkasa Putra&nik=2410511045 **Send**

Docs | Params | Authorization | Headers 7 | Body | Scripts | Settings | Cookies

Query Params

Key	Value	Description	Bulk Edit
page	1		
per_page	10		
name	Angkasa Putra		
nik	2410511045		
Key	Value	Description	

Body | Cookies | Headers 12 | Test Results

200 OK · 3.79 s · 428 B | Save Response

JSON | Preview | Visualize


```
{
    "data": [],
    "meta": {
        "page": 1,
        "per_page": 10,
        "total": 0,
        "last_page": 0
    }
}
```


uts-pplos-a-2410511045 > Patient > **Tambah Pasien**

Save | Share

POST http://localhost:3000/patients **Send**

Docs | Params | Authorization | Headers 9 | Body | Scripts | Settings | Cookies

Auth Type: Bearer Token

Token: VOR9xKQB54X9E4fq-JXxkfE_27xvls54

The authorization header will be automatically generated when you send the request. Learn more about Bearer Token authorization.

Body | Cookies | Headers 12 | Test Results

201 Created · 3.04 s · 630 B | Save Response

JSON | Preview | Visualize


```
{
    "message": "Patient created",
    "data": {
        "id": "2",
        "user_id": "6",
        "nik": "2410511046732830",
        "name": "Nadia Puspita",
        "birth_date": "2006-07-07",
        "gender": "P",
        "phone": "085813216671",
        "address": "Pamulang",
        "created_at": "2026-05-02 18:27:28",
        "updated_at": "2026-05-02 18:27:28"
    }
}
```


uts-pplos-a-2410511045 > Patient > **Lihat Detail Pasien**

Save | Share

GET http://localhost:3000/patients/2 **Send**

Docs | Params | Authorization | Headers 7 | Body | Scripts | Settings | Cookies

Auth Type: Bearer Token

Token: VOR9xKQB54X9E4fq-JXxkfE_27xvls54

The authorization header will be automatically generated when you send the request. Learn more about Bearer Token authorization.

Body | Cookies | Headers 12 | Test Results

200 OK · 1.91 s · 596 B | Save Response

JSON | Preview | Visualize


```
{
    "data": {
        "id": "2",
        "user_id": "6",
        "nik": "2410511046732830",
        "name": "Nadia Puspita",
        "birth_date": "2006-07-07",
        "gender": "P",
        "phone": "085813216671",
        "address": "Pamulang",
        "created_at": "2026-05-02 18:27:28",
        "updated_at": "2026-05-02 18:27:28"
    }
}
```

uts-pplos-a-2410511045 > Patient > **Buat Riwayat Pasien**

Save | Share

POST http://localhost:3000/records Send

Docs | Params | Authorization | Headers 9 | Body | Scripts | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

```
1  {
2      "patient_id": 2,
3      "doctor_name": "dr. Rania Putri",
4      "diagnosis": "Tipes",
5      "treatment": "Rawat Inap",
6      "visit_date": "2026-05-02"
7  }
```

Body | Cookies | Headers 12 | Test Results | 201 Created | 1.59 s | 652 B | Save Response

{} JSON | Preview | Visualize

```
1  {
2      "message": "Medical record created",
3      "data": {
4          "id": "2",
5          "patient_id": "2",
6          "doctor_name": "dr. Rania Putri",
7          "diagnosis": "Tipes",
8          "treatment": "Rawat Inap",
9          "visit_date": "2026-05-02",
10         "created_at": "2026-05-02 18:28:54",
11         "patient_name": "Nadia Puspita",
12         "nik": "2410511046732830",
13         "prescriptions": []
14     }
15 }
```

uts-pplos-a-2410511045 > Patient > **Lihat Detail Record**

Save | Share

GET http://localhost:3000/records/2 Send

Docs | Params | Authorization | Headers 7 | Body | Scripts | Settings | Cookies

Auth Type: Bearer Token

Token: VOR9xKQB54X9E4fq-JXxkfE_27xvls54

The authorization header will be automatically generated when you send the request. Learn more about Bearer Token authorization.

Body | Cookies | Headers 12 | Test Results | 200 OK | 1.60 s | 610 B | Save Response

{} JSON | Preview | Visualize

```
1  {
2      "data": {
3          "id": "2",
4          "patient_id": "2",
5          "doctor_name": "dr. Rania Putri",
6          "diagnosis": "Tipes",
7          "treatment": "Rawat Inap",
8          "visit_date": "2026-05-02",
9          "created_at": "2026-05-02 18:28:54",
10         "patient_name": "Nadia Puspita",
11         "nik": "2410511046732830",
12         "prescriptions": []
13     }
14 }
```

uts-pplos-a-2410511045 > Patient > **Prescriptions**

Save | Share

POST http://localhost:3000/prescriptions Send

Docs | Params | Authorization | Headers 9 | Body | Scripts | Settings | Cookies

none | form-data | x-www-form-urlencoded | raw | binary | GraphQL | JSON | Schema | Beautify

```
{
    "record_id": 2,
    "medicine_id": 2,
    "dosage": "500mg",
    "instruction": "3x sehari setelah makan",
    "quantity": 10
}
```

Body | Cookies | Headers 12 | Test Results

201 Created · 1.86 s · 608 B | Save Response

{} JSON | Preview | Visualize

```
{
    "message": "Prescription created",
    "data": {
        "id": "2",
        "record_id": "2",
        "medicine_id": "2",
        "dosage": "500mg",
        "instruction": "3x sehari setelah makan",
        "quantity": "10",
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